

TurboSens

An interactive turbofan benchmark for world-model evaluation

NeurIPS 2026 — Datasets & Benchmarks track

Status — multi-seed JEPA + AR-LSTM sweeps complete; counterfactual fidelity landed; per-archetype OOD discovered; raw-sensor baseline result is the surprise.

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The benchmark — what TurboSens ships

TurboSens3 + 4

Two scenarios released

Scenario 3: 400 eps; Scenario 4: 500 train + 60 + 60

7.0M flights

Total observation steps

7 sensors × 16 flight contexts, 12 stochastic events

K=10 + Kc=5

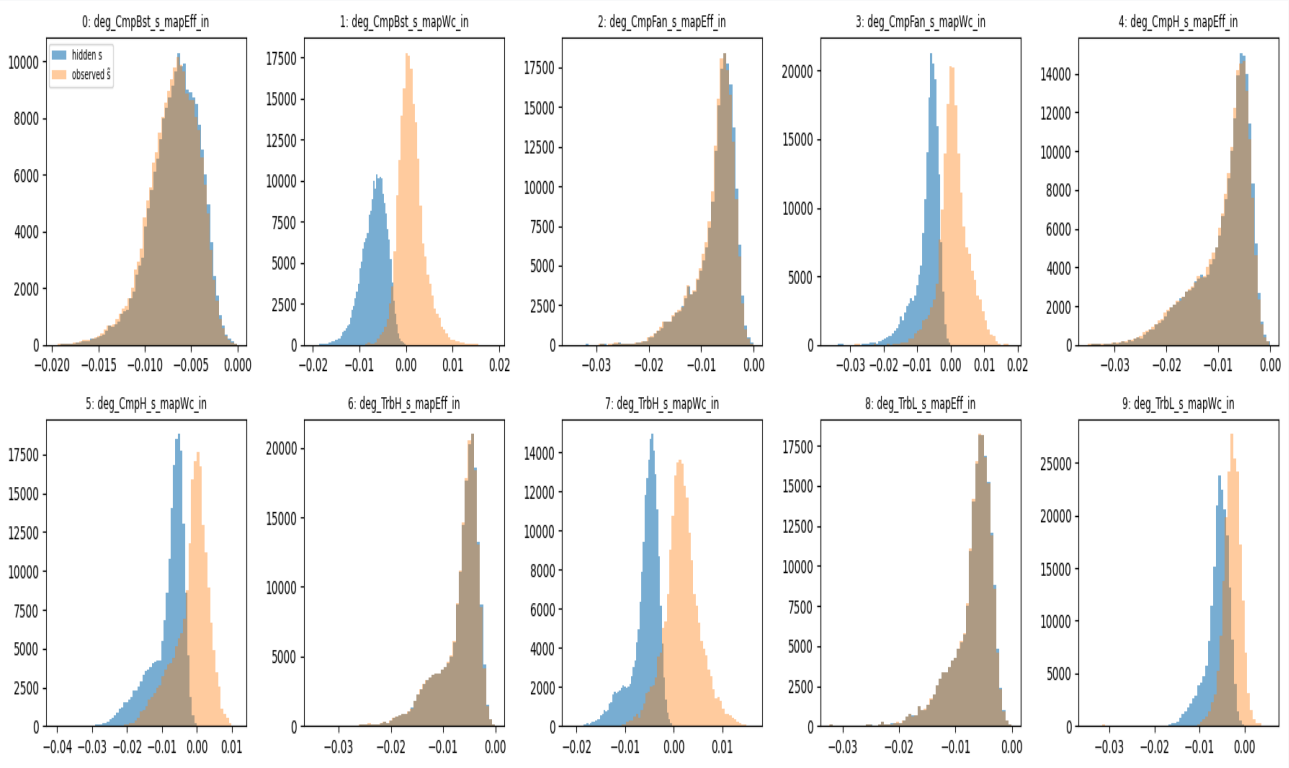
Mechanical wear + contamination

Two-channel hidden state, six operating regions

Archetype OOD

test_hard built by archetype holdout

B held out from test, A held out from test_hard



Per-component marginals: hidden s (blue) vs phantom $\tilde{s}$ (orange). Wc channels (odd indices) shifted upward by contamination — direct empirical signature of the C-c coupling.

The protocol — inverse evaluation

Train a world model self supervised on the sensor stream of the train split. Standard self supervised pretext, no access to the ground truth state.

Freeze the encoder. The world model is a black box; only the mapping $o \rightarrow z$ is exposed for downstream evaluation.

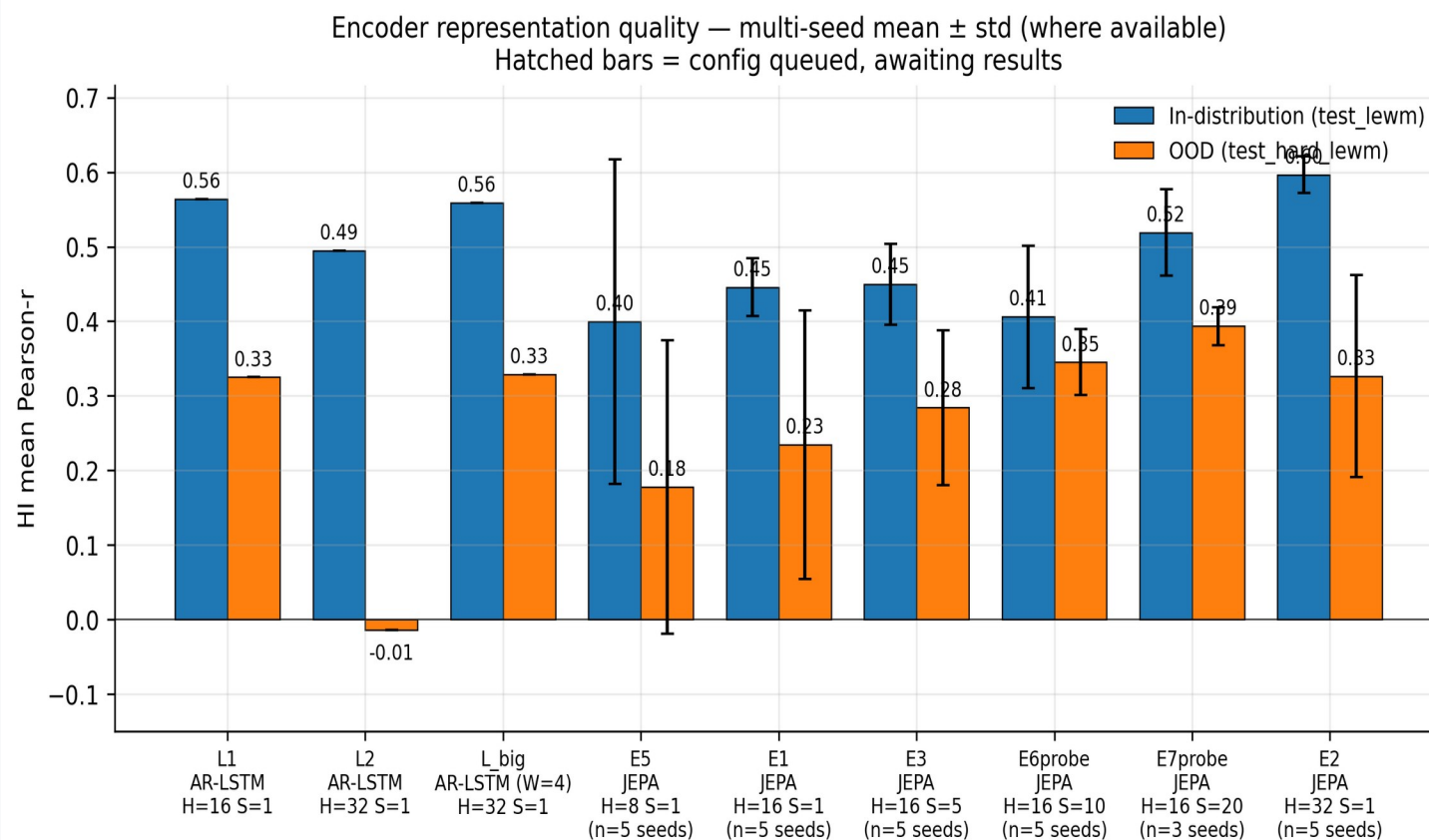
Probe the embedding against the simulator's ground truth s_t with a fixed linear regressor on the full $\sim 770k$ step train embedding stream. Report Pearson, RMSE, R^2 per HI dimension. Multi seed (mean $\pm$ std).

Evaluate on two splits: `test` (B held out) and `test_hard` (A held out). Plus a counterfactual rollout against the simulator via `EpisodeReplayer`.

Why this is novel: the simulator emits ground-truth s for every step, so we can compare against any baseline that maps observation $\rightarrow s$ — including a probe trained directly on raw sensors. World-model probes on real industrial data cannot do this.

Finding A — longer stride wins OOD

MAIN BODY



Mean HI Pearson, in-dist (blue) vs OOD (orange); error bars are multi-seed std (5–6 seeds for JEPA, single seed for AR-LSTM).

S=1 $\rightarrow$ 5 $\rightarrow$ 10 $\rightarrow$ 20 monotonic on OOD

JEPA OOD Pearson at H=16: 0.234 $\rightarrow$ 0.284 $\rightarrow$ 0.345 $\rightarrow$ 0.394.
Trend is clean across 5–6 seeds.

E7 (H=16, S=20) is the new champion

0.519 $\pm$ 0.058 in-dist, 0.394 $\pm$ 0.025 OOD — best OOD with the tightest std of any cell.

Multi-seed reranks the leaderboard

Single-seed E1 had been on top at 0.56. Multi-seed E2 leads in-dist (0.597 $\pm$ 0.025); E1 drops to 0.45.

AR-LSTM is preliminary

L1 / L2 / L_big single-seed; multi-seed rerun queued. L_big (H=32, W=4) recovers L2's OOD collapse — TemporalAggregator helps.

Finding B — raw sensors win OOD

The setup

Identical eval pipeline (770k probe-train windows, same TransformerProbe head). The *only* difference: probe input is the raw 176-dim observation tensor (7 sensors + 4 context params × 16 contexts) instead of the SSL encoder's 64-dim embedding.

OOD HI Pearson (test_hard, sl=1)

encoder	OOD Pearson	n seeds
raw_sensors (no encoder)	0.506 ± 0.043	4/5
E7 (JEPA, H=16 S=20) — best trained	0.394 ± 0.025	3/3
E6 (JEPA, H=16 S=10)	0.345 ± 0.044	5/6
L1 (AR-LSTM, H=16 S=1)	0.117 ± 0.142	4/5

What this tells us

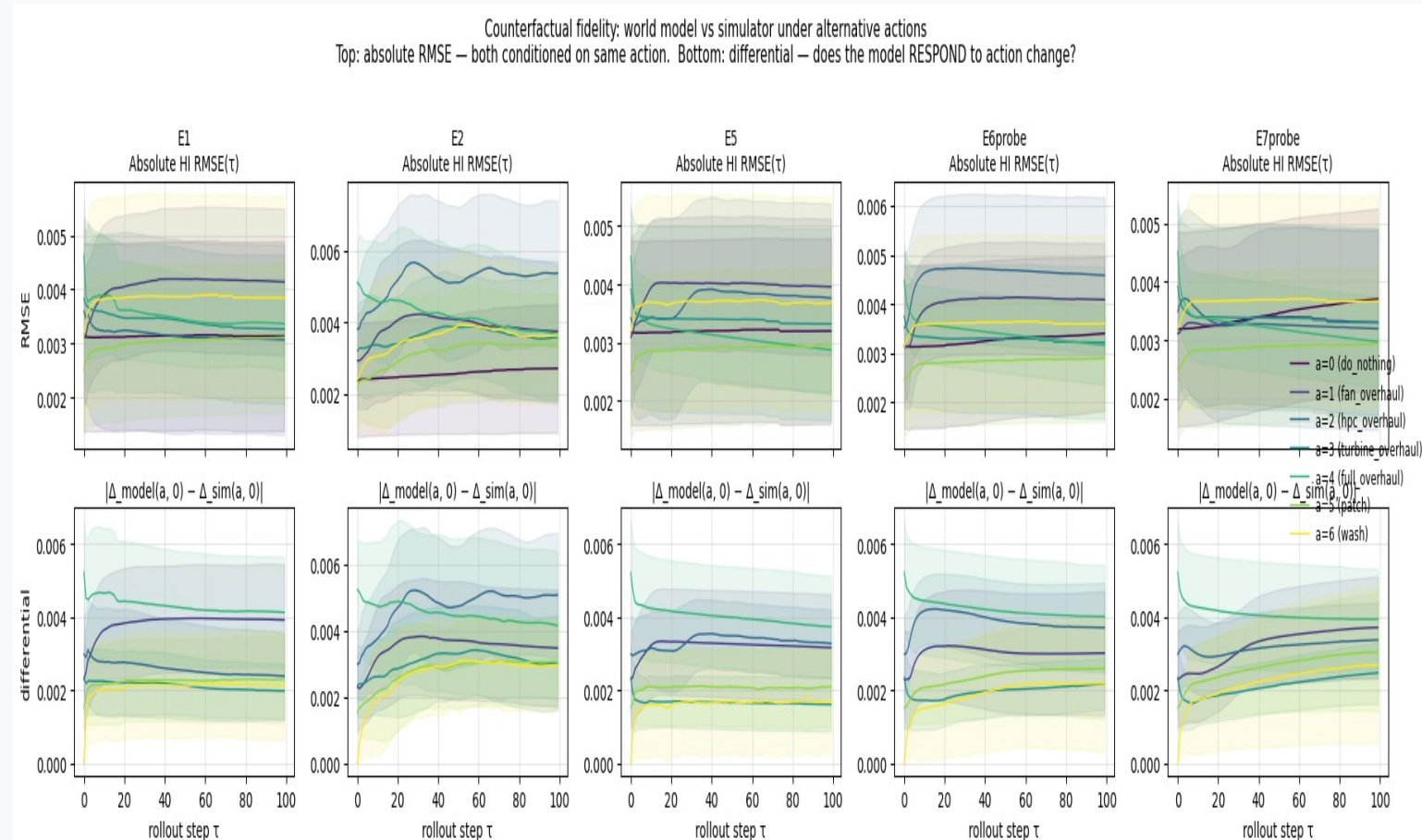
The OOD-relevant signal is in the observation. The SSL encoder compresses 176 → 64 dims and the pretext objective doesn't pressure it to keep the OOD-discriminative components.

Doesn't break the protocol — it sharpens it. The raw-sensor probe is a diagnostic only; in deployment, ground-truth s isn't available, so this baseline is unreachable in practice.

Open question for future work: how to design pretext tasks (or latent-geometry constraints, or temporal regularisers) that keep OOD-relevant information after the bottleneck.

Finding C — counterfactual fidelity

MAIN BODY



Top: absolute RMSE — model vs sim, both rolled out under the same overridden action. Bottom: differential — does the model respond to the action change like the sim does?

The setup

Branch any episode at branch_t via EpisodeReplayer; replace the recorded action with action a for τ steps; compare model's HI rollout vs sim's. 5 ckpts $\times$ 7 actions $\times$ 30 episodes $\times$ 3 branches.

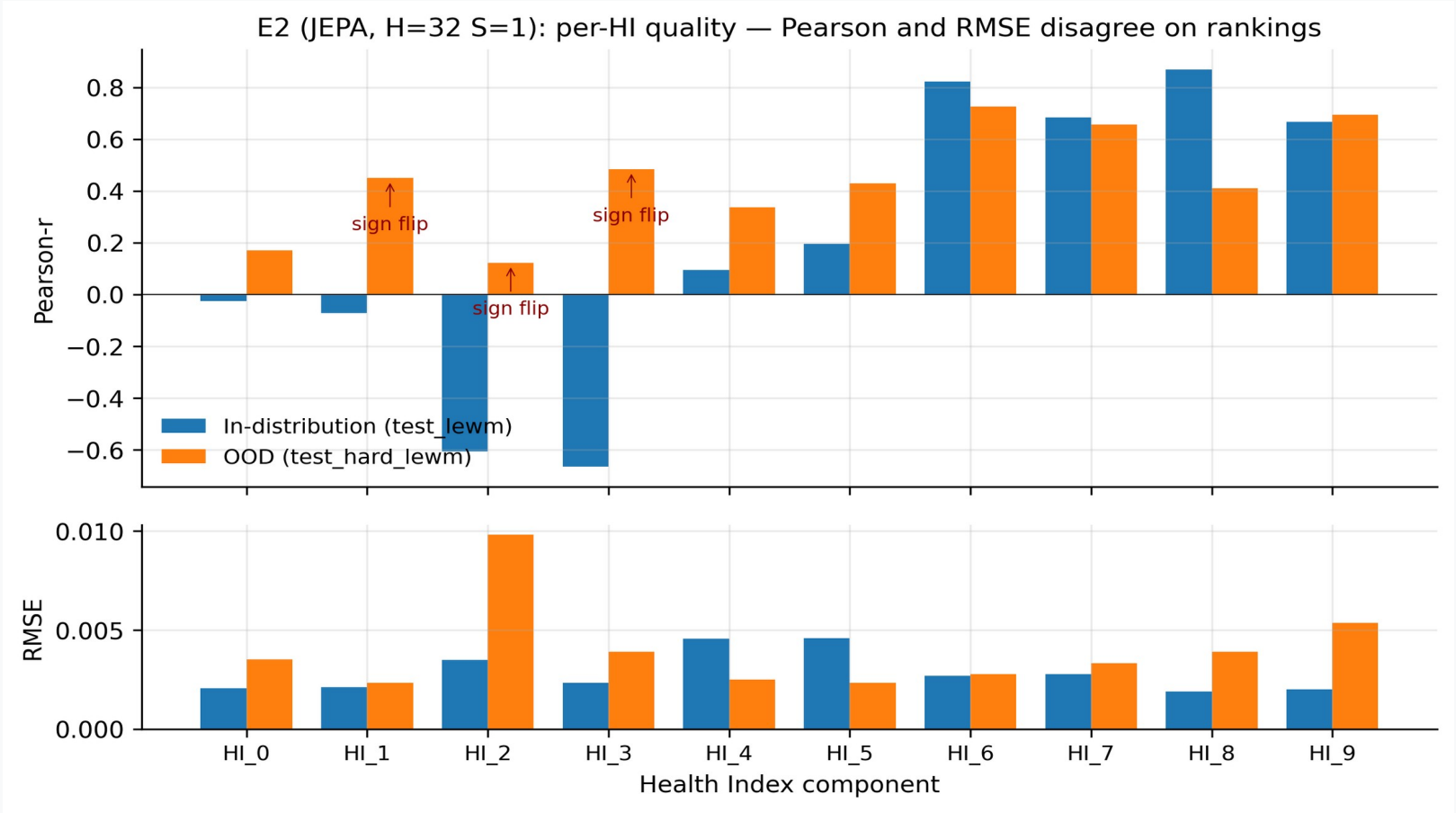
What it tells us

Models track sim within ~ 0.003 – 0.005 RMSE under counterfactual actions — fidelity is bounded.

Differential is small but non-zero — predictors do respond to action input.

(H, S) doesn't differentiate fidelity — the within-JEPA tradeoff at probe level flattens at predictor level.

Finding D — per-HI Pearson vs RMSE



What it tells us

Three sign flips on E2 (HI_1, HI_2, HI_3): the in-dist probe is anti-correlated with the ground truth on OOD for these dimensions.

Pearson and RMSE rank components differently. HI_2 has acceptable Pearson but the worst RMSE on OOD — single-metric reporting hides this.

Worst-dim > flat mean. We recommend the worst-dimension as a more conservative summary statistic.

E2 (in-dist leader): per-HI Pearson (top) and RMSE (bottom). Three sign flips on OOD; HI_2 has the worst RMSE.

Status — experiments and where they go

Main body

- Multi-seed JEPA sweep (E1-E7): the S→OOD finding (fig1, fig4).
- Raw-sensor baseline beats trained encoders OOD (sanity baselines, multi-seed).
- Per-archetype HI breakdown on L1: surfaces that test/test_hard hold out distinct archetypes.
- Per-component fig (fig2, E2): Pearson + RMSE disagreement story.
- Counterfactual fidelity (fig5): online action-conditioned rollout against the simulator.
- Cross-architecture preliminary (L1, L_big single-seed) with explicit caveat.

Appendix / deferred

- Probe-data starvation: 28k → 770k subsample → +2.3× Pearson. Methodology footnote.
- Random-encoder baseline: retracted; the 0.629 single-seed number was an outlier (multi-seed → NaN).
- E2 epoch trajectory (ep5 vs ep10): training-time specificity-vs-generality.
- Forecasting (Task 3, horizon=200) on E2 / E6 / E7: rollout RMSE + action-divergence.
- AR-LSTM multi-seed: pending OT rerun (queued).
- Per-HI multi-seed Pearson + RMSE on E2 / E6: pending dragon SUMMARY extension.
- Action-RUL probe: deferred (Pearson ≈ 0 universally; class imbalance).